Problem solving with estimation and rounding

1 A degree of accuracy shows how precise a *number* or measurement is. For example, to the nearest cm, nearest 10, 1 s.f. and so on. Fill in the blank

2 Alex buys paper £4.95, USB £□, stamps £5.20, crisps £0.95, staples £1.10 and plug £1.99. He pays with a £20 note and receives about £2 back in change. Estimate the cost of the USB. Tick **1** correct answer

- [ ] £1
- [ ] £2
- [x] £4
- [ ] £6
- [ ] £12

3 An error interval for a number y shows the range of possible values of y. It is written as: Tick **1** correct answer

- [ ] $a < y < b$
- [ ] $a > y < b$
- [ ] $a \geq y < b$
- [x] $a \leq y < b$

4 When a number, x, is rounded to the nearest integer the result is 88. Identify the error interval. Tick **1** correct answer

- [ ] $87.5 < x < 88.4\dot{9}$
- [ ] $87.5 > x < 88.4\dot{9}$
- [ ] $87.5 \leq x < 88.4\dot{9}$
- [x] $87.5 \leq x < 88.5$
- [ ] $87.5 \geq x < 88.4\dot{9}$

5 A triangle has lengths 3.5 cm (2 s.f), 4.1 cm (2 s.f) and 4.8 cm (2 s.f). Write the error interval for the perimeter p of the triangle. Tick **1** correct answer

- [x] $12.25 \leq p < 12.55$
- [ ] $12.2 \leq p < 12.5$
- [ ] $12.25 \leq p < 12.5599$
- [ ] $12.45 \leq p < 12.65$

6 A square has lengths in centimetres to 1 decimal place. The error interval of the area of the square is $17.2225 \leq x < 18.0625$. Work out the length of the square to 1 d.p. Tick **1** correct answer

- [ ] 3.8 cm
- [ ] 4.0 cm
- [x] 4.2 cm
- [ ] 4.4 cm
- [ ] 4.6 cm